

CS 255: Computer Architecture and Machine Level Programming

Prof. Martha Barker

Course Introduction + Terminal Basics

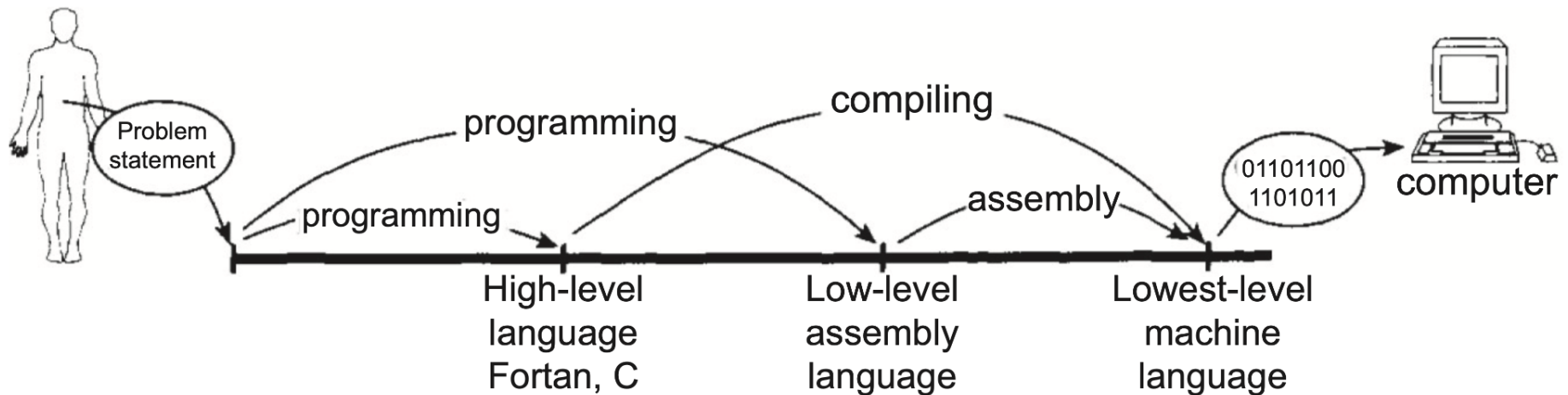
This Is a Course About How Computers Work

We will answer questions like:

What is the logical structure of a computer?

How does a computer execute code?

How does a computer store information?

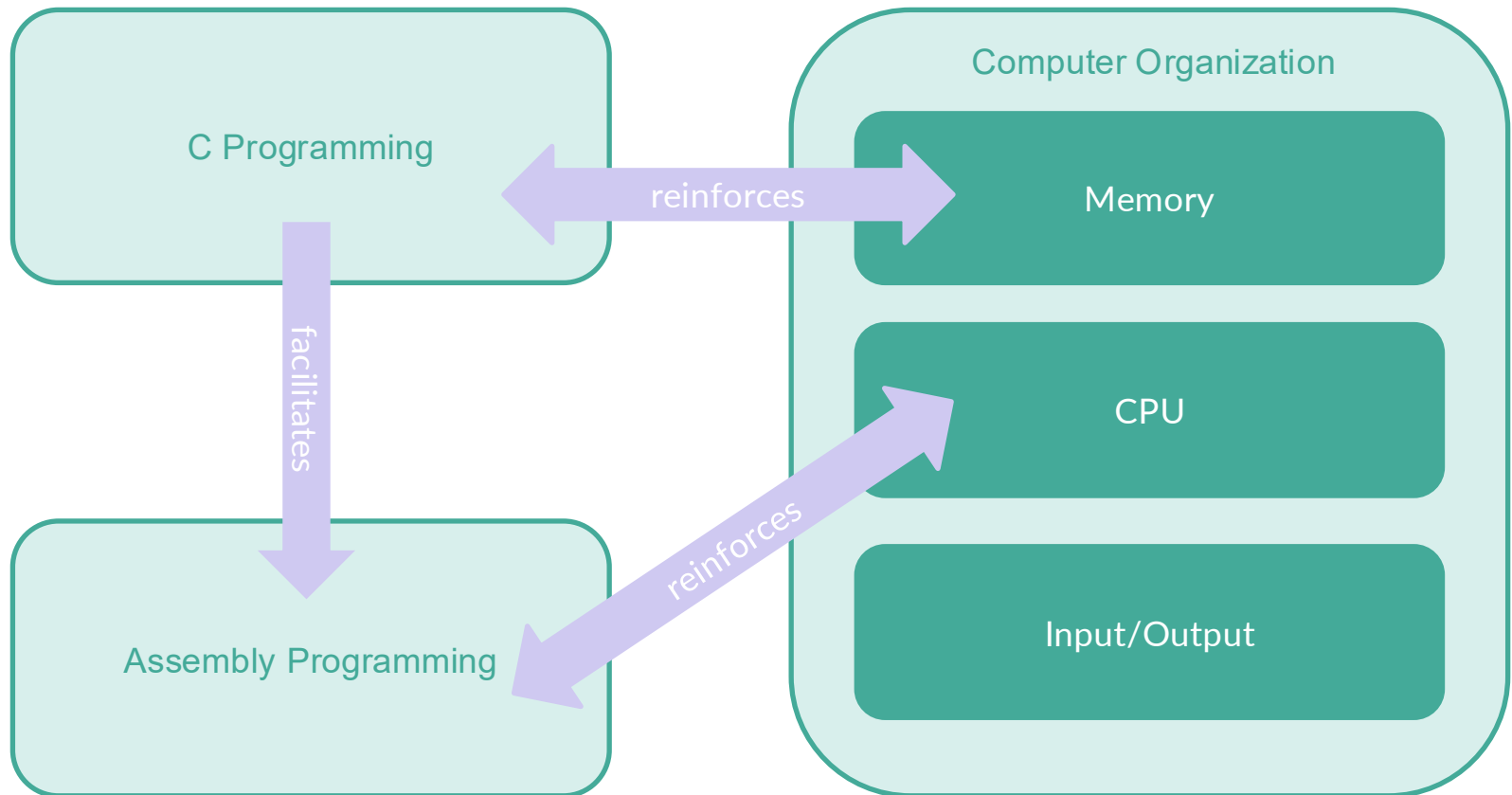


Why Does this Matter?

Understanding how code is actually executed will help you

- Improve performance
- Find and fix bugs
- Implement system software
- Understand the capabilities and limitations of computers

Components of the Course



C Programming

- Why C?
- C Basics (from Java to C)
- Compiling and Debugging C
- Pointers and Memory

Computer Organization

- What is a Computer?
- Memory and Data Representation
- Input and Output
- Computer Architecture and the CPU

Assembly Programming

- Arithmetic and Logic
- Data Movement and Memory
- Control Flow
- Subroutines and the Call Stack

Course Information

Resources

Lectures: Combination of lecture, code demos, and exercises.

Canvas: All course materials, including slides, homework assignments, and links to other sites.

Gradescope: All submissions and scoring.

Textbook: Optional reference: The C Programming Language, 2nd Edition by B. Kernighan and D. Ritchie.

Office Hours: Instructor office hours Wednesday 2-4pm MSC W302-F.
TA office hours TBA.

Grading

In-Class Exercises (5%)

Autograded Homework Assignments (20%)

- Six throughout semester
- Autograded, unlimited submissions before deadline

Written Homework Assignments (15%)

- three throughout semester
- single submission, graded once

Exams (60%)

- C Programming: February 26
- Computer Organization: March 31
- Assembly Programming: Finals week

Policies

Late Work: Homework assignments may be submitted up to three days late with a 5% per day penalty.

Regrade Requests: Must be submitted via Gradescope within one week of the assignment's return and must include a clear explanation of the grading error.

Attendance: If you cannot attend class due to medical or religious reasons, please fill out the [Student Self-Service Absence Form](#) in advance and that day's exercises will be excused.

Collaboration and AI

In-Class Exercises: You are expected to work collaboratively with your group. You may also discuss concepts with and ask questions of other groups.

AI: You may not use AI in any way to assist with in-class exercises.

Homework: You may collaborate with other students currently enrolled in CS 456. Collaboration includes discussing course content, approaches, and problems, but all submitted work must be your own.

AI: You may use AI for personal learning such as generating explanations of concepts, but you may not use AI to generate answers that you submit for homework assignments.

Staff



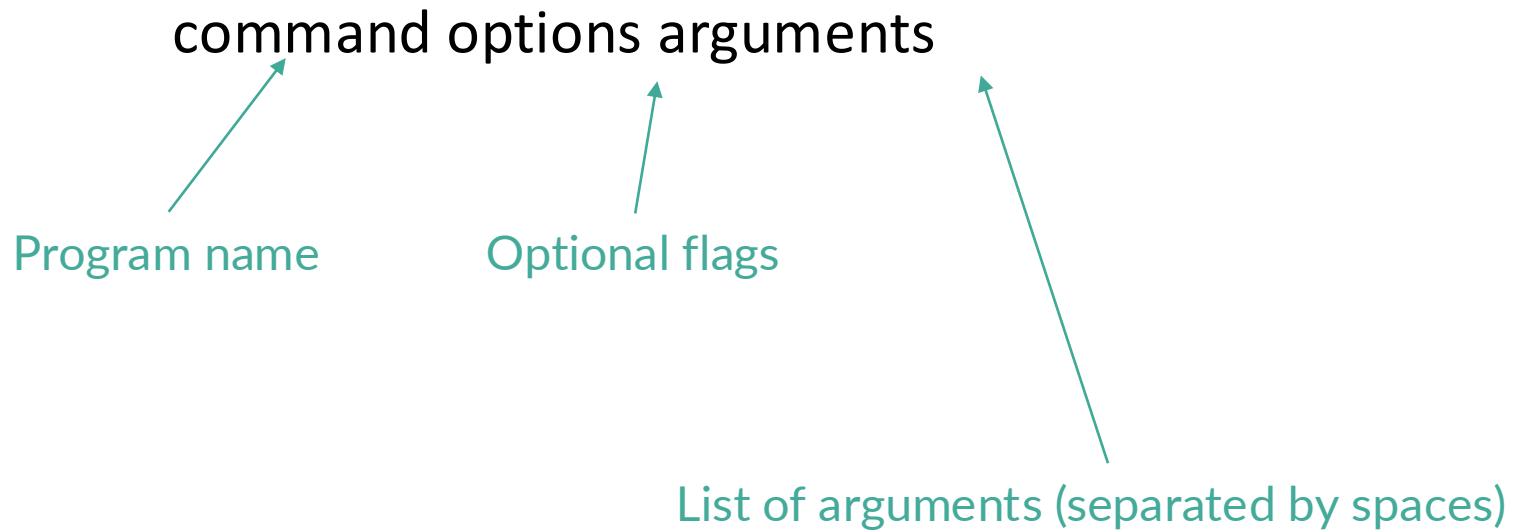
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Terminal Basics

Basic Command Structure



Change Your Default Server Terminal

To enable a more user-friendly command line interface:

1. Login to MyNetID at <https://mynetid.emory.edu>.
2. Choose 'Select UNIX or Linux Login Shell' from Selve Service options.
3. Choose '2: Bash shell (/bin/bash)' from the drop-down menu.
4. Press 'submit'. The change will take effect the next time you login to the Math/CS server.

Open a Terminal on Your Laptop

- Mac: search for 'terminal'
- Windows: search for 'cmd' or 'command prompt'

Demo: Using the Terminal

Exercise: Terminal Scavenger Hunt

Reference

Remote Access to the Server

Login to server:

```
ssh <netid>@lab0z.mathcs.emory.edu
```

Copy files to and from server:

```
sftp <netid>@<remote server>
```

```
get (download file)
```

```
put (upload file)
```

```
exit (quit sftp)
```

Navigation

Print current location: `pwd`

List everything in current directory: `ls [path]`

Change directory: `cd [path]`

File System

Create new directory: `mkdir <directory name>`

Remove a directory (must be empty): `rmdir <directory name>`

Create a file: `touch <file name>`

Copy a file: `cp <source> <destination>`

Move a file: `mv <source> <destination>`

Remove a file: `rm <file name>`

Documentation

View documentation for a command: `man <command>`

Exit documentation: `q`